

# SYDNEY TECHNICAL HIGH SCHOOL



## TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2005

# Mathematics Extension 2

### General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks - 120

- Attempt Questions 1 – 8
- All questions are of equal value

Name : \_\_\_\_\_

Teacher : \_\_\_\_\_

| Question<br>1 | Question<br>2 | Question<br>3 | Question<br>4 | Question<br>5 | Question<br>6 | Question<br>7 | Question<br>8 | Total |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
|               |               |               |               |               |               |               |               |       |

**Question 1** (15 marks)**Marks**

(a) Find  $\int \frac{2x}{x+1} dx$  2

(b) Find  $\int \frac{dx}{\sqrt{8+2x-x^2}}$  2

(c) Use partial fractions to find  $\int \frac{2}{x^2-x} dx$  3

(d) Find  $\int \sin 2x \cos^3 x \, dx$  4

(e) Find  $\int \frac{1}{(36+x^2)^{\frac{3}{2}}} dx$  using the substitution  $x = 6 \tan \theta$  4

**Question 2** (15 marks) (Start a new page)**Marks**

(a) Find the gradient of the curve  $2x^3 - x^2y + y^3 = 1$  3  
at the point  $(2, -3)$ .

(b) Solve  $|x-2| + |x+1| = 3$  2

QUESTION 2 (Continued)

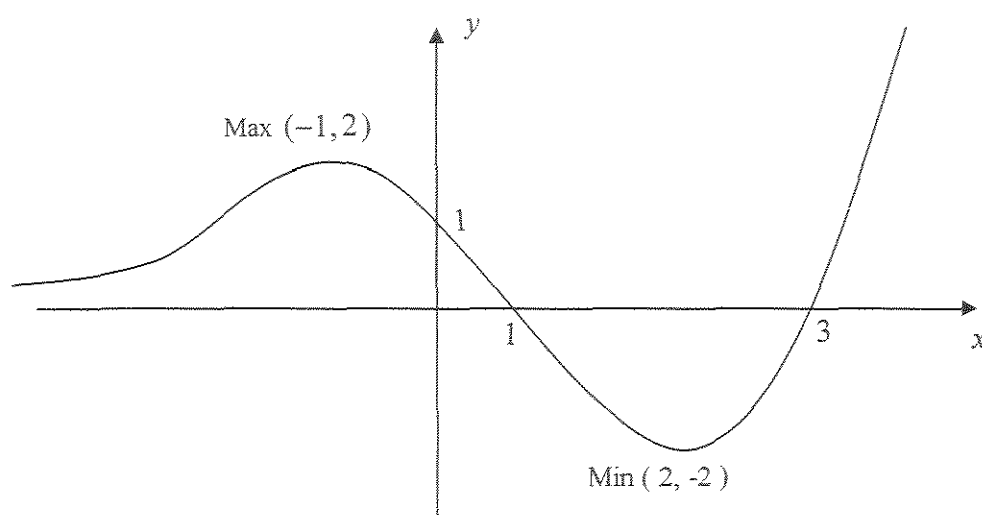
- (c) The base of a solid is the area enclosed by the curve  $y = x^2$ ,

4

the line  $x = 2$  and the  $x$  axis. Each cross-section of the solid by a plane perpendicular to the  $x$  axis is a regular hexagon with one side in the base of the solid.

Find the volume of the solid.

- (d)



The sketch above shows the function  $y = f(x)$ .

Sketch possible graphs of the following

- |                          |   |
|--------------------------|---|
| (i) $y = \frac{1}{f(x)}$ | 2 |
| (ii) $y = \int f(x) dx$  | 2 |
| (iii) $y^2 = f(x)$       | 2 |

**Question 3** (15 marks) ( Start a new page )

**Marks**

(a) Given  $z = -3\sqrt{3} + 3i$

(i) express  $z$  in modulus argument form. 2

(ii) find the smallest positive integer  $n$  such that  $z^n$  is real. 1

(b) Evaluate  $\operatorname{Im}\left(\frac{4}{1-i}\right)$  2

(c) Sketch the locus described by  $|z+2| = |z-4i|$  2

(d) (i) Sketch the intersection of the locus described by 3

$$|z| \leq 3 \text{ and } -\frac{\pi}{4} \leq \arg(z+3) \leq \frac{\pi}{4}$$

(ii) If the complex number  $\omega$  lies on the boundary of the region  
sketched in part (i), find the minimum value of  $|\omega|$ . 2

(e) OABC is a rectangle on the Argand diagram in which side OC  
is twice the length of OA, where O is the origin.

(i) If A represents the complex number  $1+2i$ , find the complex numbers 2  
represented by B and C given that the argument of the complex number  
represented by the point C is negative..

(ii) If this rectangle is rotated anticlockwise  $\frac{\pi}{3}$  radians about O, find 1  
the complex number represented by the new position of A.

**Question 4** (15 marks) ( Start a new page )

**Marks**

(a) For the hyperbola with equation  $4x^2 - 9y^2 = 36$  find,

(i) the eccentricity 2

(ii) the equation of the asymptotes 1

(b) Given the point  $P(a \sec \theta, b \tan \theta)$  lies on the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(i) Show that the tangent to the hyperbola  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

at the point  $P(a \sec \theta, b \tan \theta)$  has equation  $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$  3

(ii) If this tangent in part (i) meets the directrix of the hyperbola 4

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  corresponding to the focus at  $S(ae, 0)$  at the point  $Q$ ,

show that  $\angle PSQ$  is a right angle.

(c) (i) Show that  $(1 - \sqrt{x})^{n-1} \sqrt{x} = (1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n$  1

(ii) If  $I_n = \int_0^1 (1 - \sqrt{x})^n dx$  for  $n \geq 0$  4

show that  $I_n = \frac{n}{n+2} I_{n-1}$  for  $n \geq 1$ .

**Question 5** (15 marks) ( Start a new page )

**Marks**

- (a) The region bounded by the curves  $y = x^2$  and  $y = x + 2$

4

is rotated about the line  $x = 3$ .

Use the method of cylindrical shells to find the volume of the solid of revolution formed.

- (b) Solve the equation  $8x^4 + 12x^3 - 30x^2 + 17x - 3 = 0$

4

given that it has a triple root.

- (c) If  $\alpha, \beta, \delta$  are the roots of  $x^3 + px^2 + qx + r = 0$ ,

2

find the polynomial equation with roots  $\alpha^2, \beta^2, \delta^2$ .

- (d) The acceleration of a particle moving in simple harmonic motion

is given by  $\ddot{x} = -n^2x$  where  $x$  is the displacement of the particle from the origin and  $n$  is a constant.

- (i) Show that the velocity  $v$  of the particle is given by

2

$$v^2 = n^2(a^2 - x^2) \quad \text{where } a \text{ is the amplitude of the motion.}$$

- (ii) Given that the speed of the particle is  $V \text{ m/s}$  when it is  $d$  metres

from the origin and that its speed is  $\frac{V}{2} \text{ m/s}$  when it is  $2d$  metres

from the origin, show that :

$\alpha$ ) the particle's amplitude is  $\sqrt{5}d$  metres.

2

$\beta$ ) the period of the motion is  $\frac{4\pi d}{V}$  seconds.

1

**Question 6** (15 marks) ( Start a new page )

**Marks**

(a) Evaluate  $\int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$  using the substitution  $t = \tan \frac{x}{2}$

4

(b) A particle of mass  $m$  is fired vertically upwards with initial velocity  $V$  m/s and is subjected to air resistance equal to  $mkv$  Newtons where  $k$  is a constant and  $v$  is the velocity of the particle in metres per second as it moves through the air.

(i) Explain why the equation of motion of the particle is given by

1

$$\ddot{x} = -g - kv \quad \text{where } g \text{ is the acceleration due to gravity.}$$

(ii) Show that the maximum height reached by the particle is given by

4

$$H = \frac{V}{k} - \frac{g}{k^2} \ln \left( 1 + \frac{kV}{g} \right)$$

(c) (i) Find the seven complex roots of the equation  $z^7 = 1$ .

2

(ii) If  $\omega$  is the complex root of  $z^7 = 1$

1

with smallest positive argument, find the value of

$$1 + \omega + \omega^2 + \omega^3 + \omega^4 + \omega^5 + \omega^6$$

(iii) Find the cubic equation whose roots are

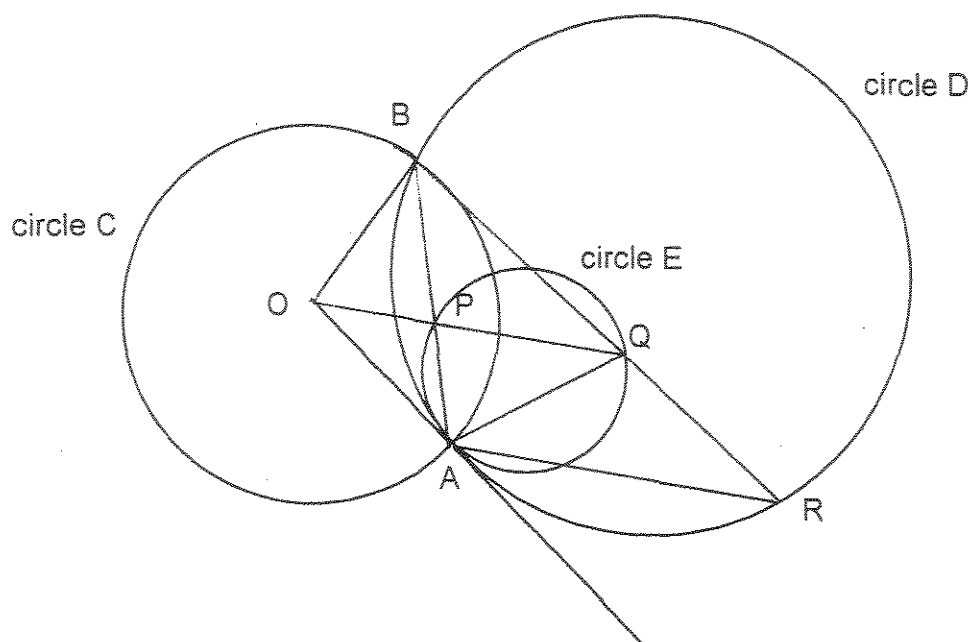
3

$$\omega + \omega^6, \quad \omega^2 + \omega^5, \quad \omega^3 + \omega^4$$

**Question 7** (15 marks) (Start a new page)

Marks

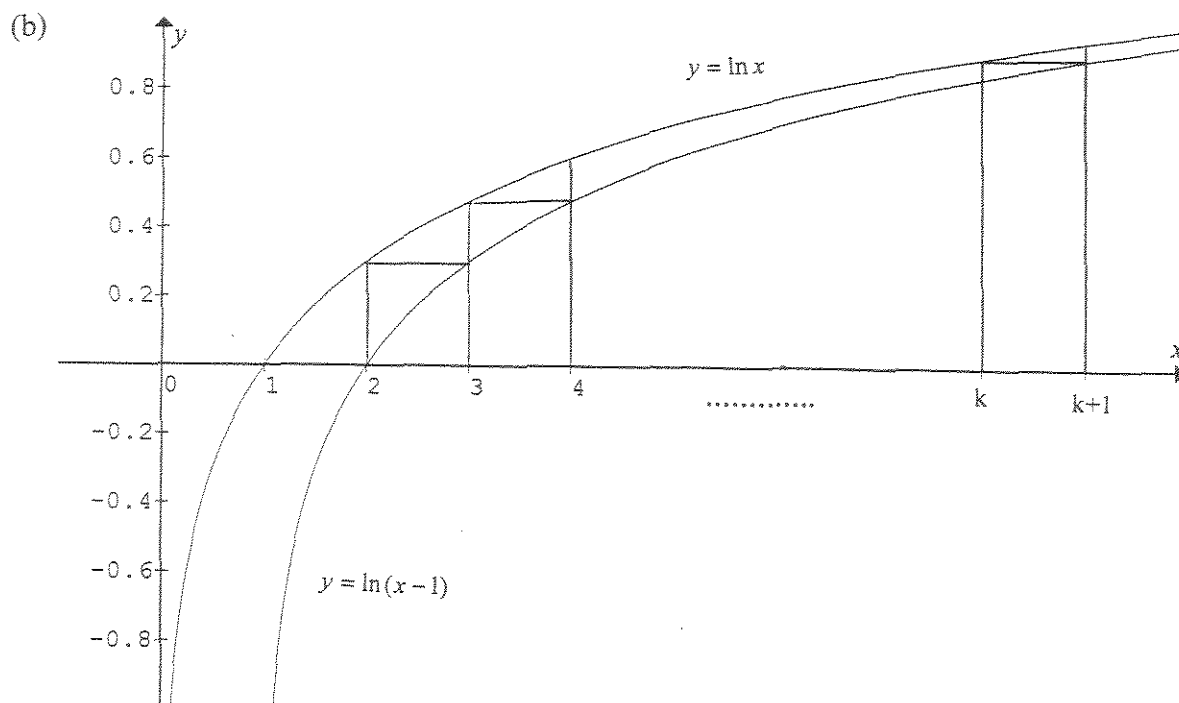
(a)



In the diagram above,  $OA$  is a radius of a circle  $C$  with centre  $O$ , and two circles  $D$  and  $E$  are drawn touching the line  $OA$  at  $A$  as shown. The larger circle  $D$  meets circle  $C$  again at  $B$ , and the line  $AB$  meets the smaller circle  $E$  again at  $P$ . The line  $OP$  meets circle  $E$  again at  $Q$ , and the line  $BQ$  meets the circle  $D$  again at  $R$ .

- |                                                                     |   |
|---------------------------------------------------------------------|---|
| (i) Let $\angle OAP = \theta$ . Explain why $\angle PQA = \theta$ . | 1 |
| (ii) Prove that the points $O$ , $B$ , $Q$ and $A$ are concyclic.   | 2 |
| (iii) Prove that $OQ$ bisects $\angle BQA$ .                        | 2 |
| (iv) Prove that $OQ \parallel AR$                                   | 2 |

QUESTION 7 (Continued)



In the diagram above, the curves  $y = \ln x$  and  $y = \ln(x-1)$  are sketched and  $k-1$  rectangles are constructed between  $x=2$  and  $x=k+1$  where  $k \geq 2$ .  
Let  $S = \ln 2 + \ln 3 + \ln 4 + \dots + \ln k$ .

(i) Explain why  $S$  represents the sum of the areas of the  $k-1$  rectangles. 1

(ii) Use an appropriate integration method to show that

$$\int_2^{k+1} \ln(x-1) dx = k \ln k - k + 1 \quad 4$$

(iii) Hence show that  $k^k < k!e^{k-1} < \frac{1}{4}(k+1)^{k+1}$  where  $k \geq 2$  3

( note  $n! = n(n-1)(n-2)\dots\dots\dots 3 \times 2 \times 1$  )

**Question 8** (15 marks) ( Start a new page )

**Marks**

(a) Find all  $x$  such that  $\sin x = \cos 5x$  and  $0 < x < \pi$ .

3

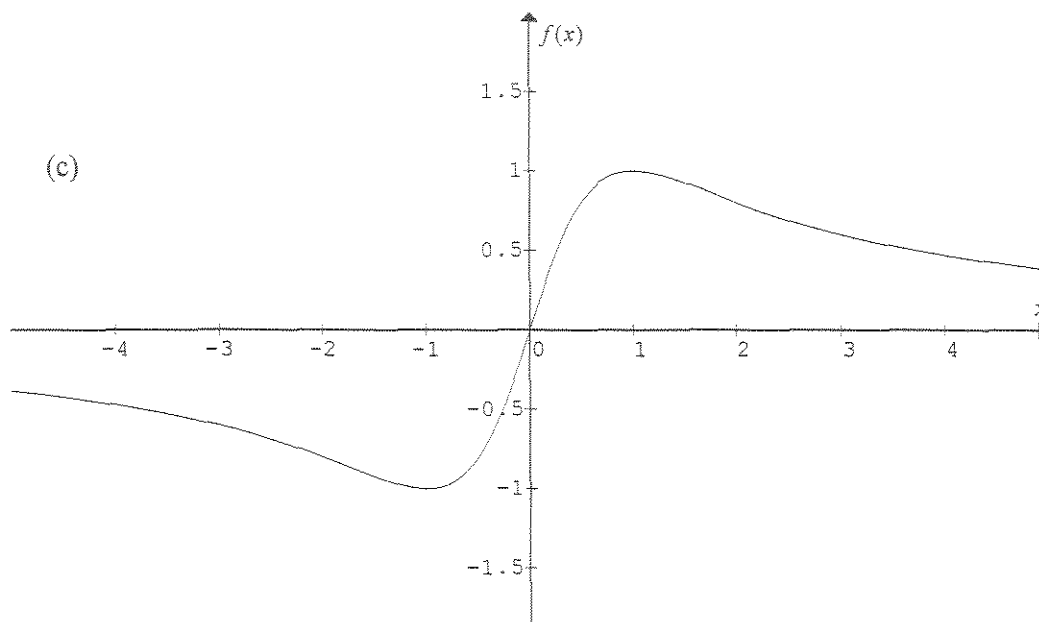
(b) If  $z$  is a complex number for which

2

$$|z| = 1 \text{ and } \arg(z) = \theta, \quad 0 \leq \theta \leq \frac{\pi}{2},$$

find the value of  $\arg\left(\frac{2}{1-z^2}\right)$  in terms of  $\theta$ .

**Question 8 continued on next page.**



The curve  $f(x) = \frac{2x}{1+x^2}$  is sketched above. It has a maximum turning point at  $(1,1)$  and minimum turning point at  $(-1,-1)$ .

(i) State the range of  $f(x) = \frac{2x}{1+x^2}$  1

(ii) Let  $x_0$  be a real number not equal to 1 or -1 and consider the sequence of real numbers defined by  $x_{n+1} = f(x_n)$  for  $n = 0, 1, 2, \dots$

( $\alpha$ ) Given  $x_1 = g(r)$  and  $g(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$  express  $r$  in terms of  $x_1$ . 2

( $\beta$ ) Hence deduce that there exists a real number  $r$  such that  $x_1 = g(r)$  1  
 where  $g(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$ .

( $\delta$ ) Show that  $\frac{2g(x)}{1+(g(x))^2} = g(2x)$ . 2

( $\gamma$ ) Hence, using the above results and Mathematical Induction 4

show that  $x_n = g(2^{n-1} r)$  for  $n = 1, 2, 3, \dots$

**End of Paper**

# Solutions

Ext II 2005 Trial HSC

## QUESTION 1

$$\begin{aligned} \text{a) } \int \frac{2x}{x+1} dx \\ &= 2 \int 1 - \frac{1}{x+1} dx \\ &= 2x - 2 \ln(x+1) + c \end{aligned}$$

$$\begin{aligned} \text{b) } \int \frac{dx}{\sqrt{8+2x-x^2}} \\ &= \int \frac{dx}{\sqrt{9-(x-1)^2}} \\ &= \sin^{-1} \frac{x-1}{3} + c \end{aligned}$$

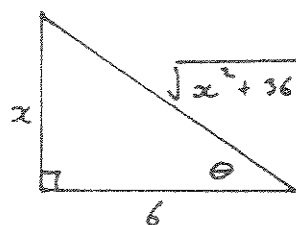
$$\begin{aligned} \text{c) } \int \frac{2}{x^2-x} dx \\ \frac{2}{x(x-1)} = \frac{A}{x} + \frac{B}{x-1} \\ \therefore 2 = A(x-1) + B(x) \\ \therefore A = -2, B = 2 \end{aligned}$$

$$\begin{aligned} \therefore \int \frac{2}{x^2-x} dx \\ &= \int \frac{2}{x-1} - \frac{2}{x} dx \\ &= 2 \ln(x-1) - 2 \ln x + c \\ &= 2 \ln \left( \frac{x-1}{x} \right) + c \end{aligned}$$

$$\begin{aligned} \text{d) } \int \sin 2x \cos^3 x dx \\ &= 2 \int \sin x \cos^4 x dx \\ &= -\frac{2}{5} \cos^5 x + c \end{aligned}$$

$$\begin{aligned} \text{e) } x &= 6 \tan \theta \\ dx &= 6 \sec^2 \theta d\theta \end{aligned}$$

$$\begin{aligned} \int \frac{1}{(36+x^2)^{\frac{3}{2}}} dx \\ &= \int \frac{6 \sec^2 \theta d\theta}{(36+36 \tan^2 \theta)^{\frac{3}{2}}} \\ &= \frac{1}{36} \int \frac{\sec^2 \theta d\theta}{(1+\tan^2 \theta)^{\frac{3}{2}}} \\ &= \frac{1}{36} \int \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^{\frac{3}{2}}} \\ &= \frac{1}{36} \int \frac{d\theta}{\sec \theta} \\ &= \frac{1}{36} \int \cos \theta d\theta \\ &= \frac{1}{36} \sin \theta + c \end{aligned}$$



$$= \frac{x}{36 \sqrt{x^2+36}} + c$$

## QUESTION 2

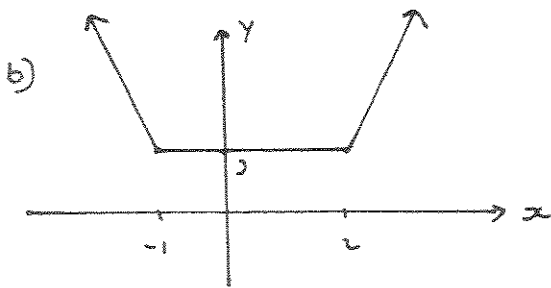
a)  $2x^3 - x^2y + y^3 = 1$

$$6x^2 - \left[ 2xy + x^2 \frac{dy}{dx} \right] + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2xy - 6x^2}{-x^2 + 3y^2}$$

when  $x=2, y=-3$

$$\begin{aligned} \therefore m_T &= \frac{-12 - 24}{-4 + 27} \\ &= \frac{-36}{23} \end{aligned}$$



$$\therefore -1 \leq x \leq 2$$

c) hexagon is 6 equilateral  $\Delta$ 's with side length  $y$ .

$$\begin{aligned} A(x) &= 6 \times \frac{1}{2} \times y \times y \times \sin 60^\circ \\ &= \frac{3\sqrt{3}}{2} y^2 \end{aligned}$$

but  $y = x^2$

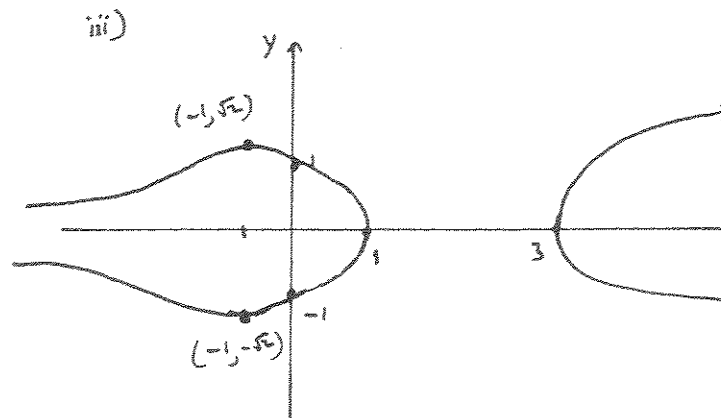
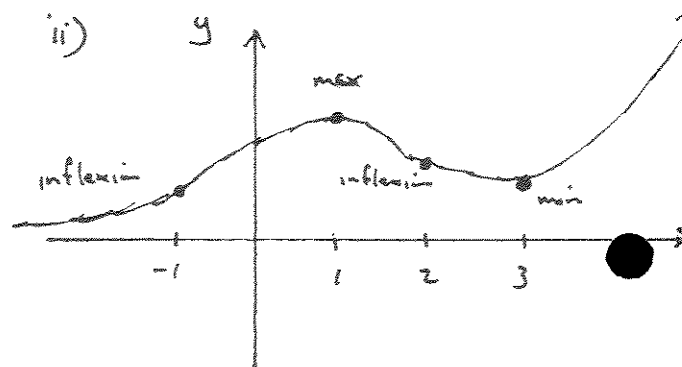
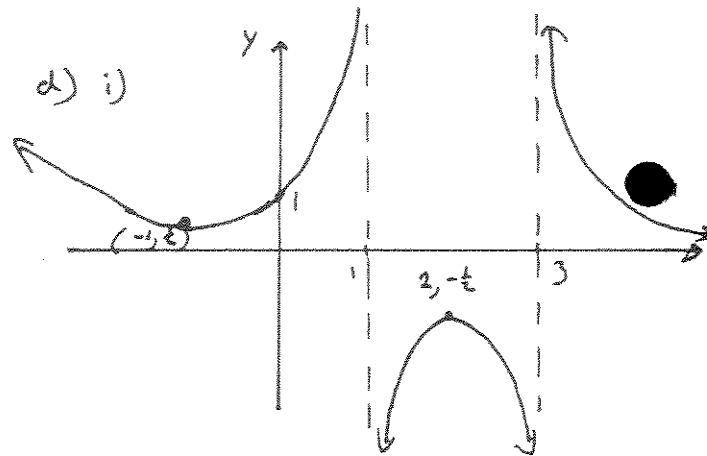
$$\therefore A(x) = \frac{3\sqrt{3}}{2} x^4$$

$$\therefore \Delta V(x) = \frac{3\sqrt{3}}{2} x^4 dx$$

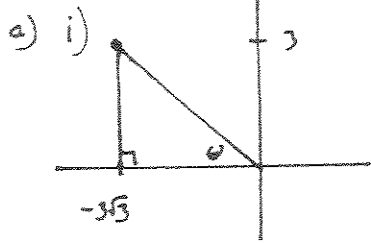
$$\therefore V = \int_0^2 \frac{3\sqrt{3}}{2} x^4 dx$$

$$= \left[ \frac{3\sqrt{3}}{10} x^5 \right]_0^2$$

$$= \frac{48\sqrt{3}}{5} \text{ cubic units}$$



### QUESTION 3



$$|z| = \sqrt{(3\sqrt{3})^2 + 3^2}$$

$$= \sqrt{36}$$

$$= 6$$

$$\tan \theta = \frac{3}{3\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

$$\therefore \arg(z) = \frac{5\pi}{6}$$

$$\therefore z = 6 \left( \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$$

ii) to be real  $\frac{5n\pi}{6} = m\pi$  ( $m$  is integer)

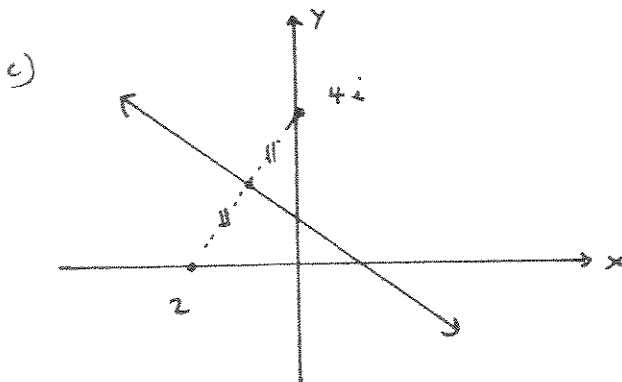
$\therefore n$  equals 6.

b)

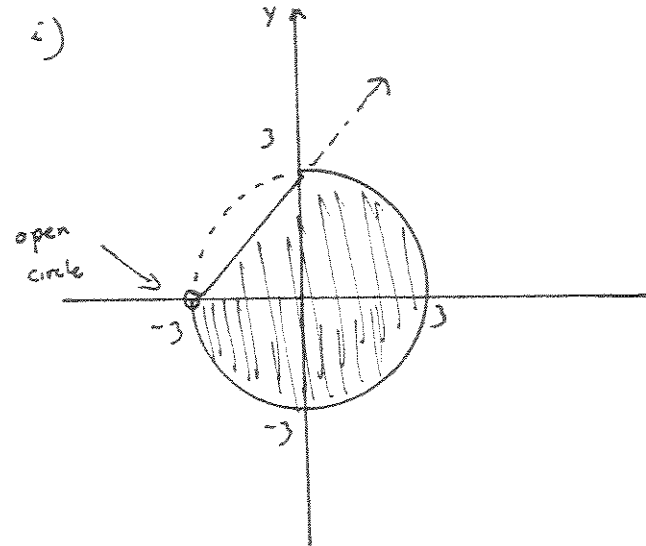
$$\frac{4}{1-i} = \frac{4}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{4+4i}{2}$$

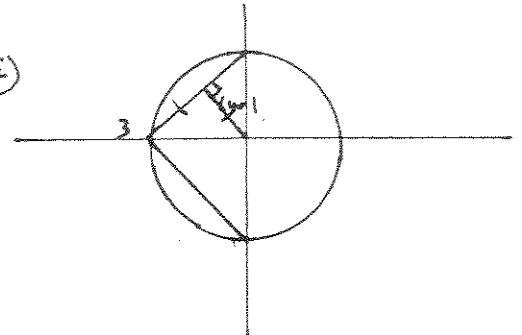
$$\therefore \operatorname{Im}\left(\frac{4}{1-i}\right) = 2$$



d) i)



ii)



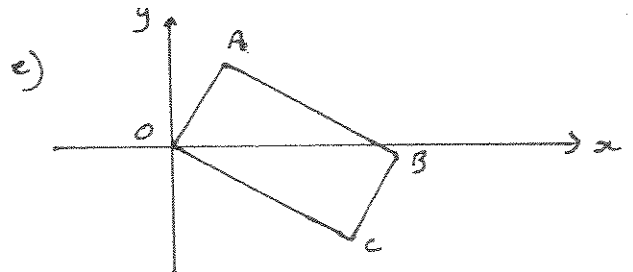
let  $|w| = x$

$$\therefore x^2 + x^2 = 3^2$$

$$2x^2 = 9$$

$$x = \frac{3}{\sqrt{2}}$$

$$\therefore \min |w| = \frac{3}{\sqrt{2}}$$



i)

$$C = (-i)(2)(1+2i)$$

$$= 4-2i$$

$$B = (1+2i) + (4-2i)$$

$$= 5$$

ii) multiply A by  $\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$   
or  $\frac{1}{2} + i \frac{\sqrt{3}}{2}$

$$\therefore A' = (1+2i)\left(\frac{1}{2} + i \frac{\sqrt{3}}{2}\right)$$

$$= \frac{1}{2} - \sqrt{3} + i\left(\frac{\sqrt{3}}{2} + 1\right)$$

#### QUESTION 4

a)  $\frac{x^2}{9} - \frac{y^2}{4} = 1$

i)  $b^2 = a^2(e^2 - 1)$   
 $4 = 9(e^2 - 1)$   
 $e = \frac{\sqrt{13}}{3}$

ii)  $y = \pm \frac{2x}{3}$

b) i)  $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\therefore m_T = \frac{b^2 a \sec \theta}{a^2 b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$\therefore$  using  $y - y_1 = m(x - x_1)$   
 $y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$

$$bx \sec \theta - ay \tan \theta = ab(\sec^2 \theta - \tan^2 \theta)$$

$$bx \sec \theta - ay \tan \theta = ab$$

$$\therefore \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

ii) directrix  $x = \frac{a}{e}$

sub. into tangent

$$\frac{a}{e} \cdot \frac{\sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$y = \frac{b(\sec \theta - e)}{e \tan \theta}$$

$$\therefore Q\left(\frac{a}{e}, \frac{b(\sec \theta - e)}{e \tan \theta}\right)$$

$$P(a \sec \theta, b \tan \theta)$$

$$S(ae, 0)$$

$$\therefore m_{QS} = \frac{\frac{b(\sec \theta - e)}{e \tan \theta}}{\frac{a}{e} - ae}$$

$$= \frac{b(\sec \theta - e)}{a(1 - e^2) \tan \theta}$$

$$m_{PS} = \frac{b \tan \theta}{a(\sec \theta - e)}$$

$$\therefore m_{QS} \times m_{PS} =$$

$$\frac{b \tan \theta}{a(\sec \theta - e)} \cdot \frac{b(\sec \theta - e)}{a(1 - e^2) \tan \theta}$$

$$= \frac{b^2}{a^2(1 - e^2)}$$

$$= \frac{b^2}{-b^2} \quad b^2 = a^2(e^2 - 1)$$

$$= -1$$

$\therefore \angle PSQ$  is right angle.

$$\begin{aligned}
 \text{c) i) RHS} &= (1-\sqrt{x})^{n+1} - (1-\sqrt{x})^n \\
 &= (1-\sqrt{x})^{n+1} [1 - (1-\sqrt{x})] \\
 &= (1-\sqrt{x})^{n+1} \sqrt{x} \\
 &= \text{LHS}
 \end{aligned}$$

$$\text{ii) } I_n = \int_0^1 (1-\sqrt{x})^n dx$$

$$\begin{aligned}
 u &= (1-\sqrt{x})^n & u' &= n(1-\sqrt{x})^{n-1} \cdot -\frac{1}{2} x^{-\frac{1}{2}} \\
 v &= x & v' &= 1
 \end{aligned}$$

$\therefore$  using integration by parts

$$\begin{aligned}
 I_n &= \left[ x(1-\sqrt{x})^n \right]_0^1 + n \int_0^1 x(1-\sqrt{x})^{n-1} \cdot \frac{1}{2} x^{-\frac{1}{2}} dx \\
 &= \frac{n}{2} \int_0^1 \sqrt{x} (1-\sqrt{x})^{n-1} dx \\
 &= \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n dx \\
 &= \frac{n}{2} [I_{n-1} - I_n]
 \end{aligned}$$

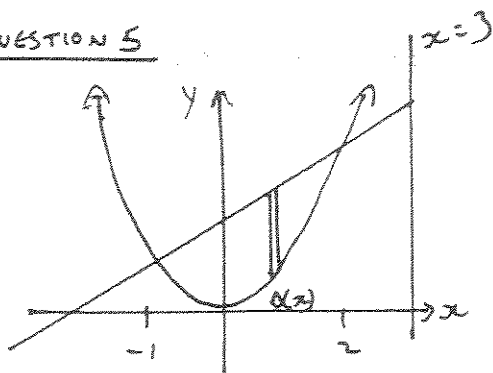
$$\left(\frac{n}{2} + 1\right) I_n = \frac{n}{2} I_{n-1}$$

$$(n+2) I_n = n I_{n-1}$$

$$I_n = \frac{n}{n+2} I_{n-1}$$

# QUESTION 5

a)



$y = x^2$  need pt. of  
 $y = x + 2$  intersection  
 by inspection  $x = -1, 2$ .

$$\begin{aligned}\Delta V &= 2\pi \times \text{radius} \times \text{height} \times \text{thickness} \\ &= 2\pi \times (3-x) \times (x+2-x^2) \times \Delta x \\ &= 2\pi [6 + x - 4x^2 + x^3] \Delta x\end{aligned}$$

$$\begin{aligned}\therefore V &= 2\pi \int_{-1}^2 (6 + x - 4x^2 + x^3) dx \\ &= 2\pi \left[ 6x + \frac{1}{2}x^2 - \frac{4}{3}x^3 + \frac{1}{4}x^4 \right]_{-1}^2 \\ &= \frac{45\pi}{2} \text{ cubic units}\end{aligned}$$

b)

$$P(x) = 8x^4 + 12x^3 - 30x^2 + 17x - 3$$

$$P'(x) = 32x^3 + 36x^2 - 60x + 17$$

$$P''(x) = 96x^2 + 72x - 60$$

a root of  $P''(x) = 0$  is triple root of  $P(x) = 0$

$$96x^2 + 72x - 60 = 0$$

$$12(8x^2 + 6x - 5) = 0$$

$$12(4x + 5)(2x - 1) = 0$$

$$x = -\frac{5}{4}, \frac{1}{2}$$

$$P'(-\frac{5}{4}) \neq 0$$

$$P'(\frac{1}{2}) = 0$$

$\therefore x = \frac{1}{2}$  is triple root

$$\therefore \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \alpha = -\frac{12}{8}$$

$$\alpha = -3$$

$$\therefore x = -3, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}$$

QUESTION 5 (cont.)

c) required polynomial is  $P(\sqrt{x}) = 0$

$$(\sqrt{x})^3 + p(\sqrt{x})^2 + q\sqrt{x} + r = 0$$

$$x\sqrt{x} + q\sqrt{x} = -px - r$$

$$(x\sqrt{x} + q\sqrt{x})^2 = (-px - r)^2$$

$$x^3 + 2qx^2 + q^2x = p^2x^2 + 2prx + r^2$$

$$x^3 + (2q - p^2)x^2 + (q^2 - 2pr)x - r^2 = 0$$

d) i)  $\ddot{x} = -n^2x$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = -n^2x$$

$$\frac{1}{2}v^2 = -\frac{n^2}{2}x^2 + c$$

when  $v=0$   $x=a$

$$\therefore c = \frac{n^2a^2}{2}$$

$$\therefore \frac{1}{2}v^2 = \frac{n^2a^2}{2} - \frac{n^2x^2}{2}$$

$$v^2 = n^2(a^2 - x^2)$$

ii) from information

$$v^2 = n^2(a^2 - d^2)$$

$$\frac{v^2}{4} = n^2(a^2 - 4d^2)$$

α) Solve for  $a$

$$\therefore \frac{n^2(a^2 - d^2)}{4} = n^2(a^2 - 4d^2)$$

$$a^2 - d^2 = 4a^2 - 16d^2$$

$$3a^2 = 15d^2$$

$$a^2 = 5d^2$$

$$a = \sqrt{5}d$$

$$\therefore \text{amplitude} = \sqrt{5}d$$

β) using part d)

$$v^2 = n^2(5d^2 - d^2)$$

$$v^2 = n^2 4d^2$$

$$n^2 = \frac{v^2}{4d^2}$$

$$n = \frac{v}{2d}$$

$$\therefore \text{period} = \frac{2\pi}{n}$$

$$= \frac{4\pi d}{v}$$

# QUESTION 6

$$a) \int_0^{\frac{\pi}{2}} \frac{dx}{2 + \cos x}$$

$$t = \tan \frac{x}{2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$\therefore \int_0^1 \frac{\frac{2dt}{1+t^2}}{2 + \frac{1-t^2}{1+t^2}}$$

$$= \int_0^1 \frac{2dt}{3+t^2}$$

$$= \left[ \frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}}$$

b) i) direction of motion +ve  
two forces ( $mg$  and  $mkv$ )  
acting against motion

$$\therefore ma = -mg - mkv$$

$$\therefore \ddot{x} = -g - kv$$

$$ii) \ddot{x} = -g - kv$$

$$\therefore v \frac{dv}{dx} = -(g + kv)$$

$$\int \frac{v dv}{g + kv} = -\int dx$$

$$\frac{1}{k} \int \frac{g + kv}{g + kv} - \frac{g}{g + kv} dv = -x + c$$

$$\frac{1}{k} \int 1 - \frac{g}{g + kv} dv = -x + c$$

$$\frac{1}{k} \left[ v - \frac{g}{k} \ln(g + kv) \right] = -x + c$$

$$\text{when } x=0 \quad v=V$$

$$\therefore c = \frac{1}{k} \left[ V - \frac{g}{k} \ln(g + kV) \right]$$

$v=0$  at greatest height ( $x=H$ )

$$\therefore \frac{1}{k} \left[ -\frac{g}{k} \ln g \right] = -H + \frac{1}{k} \left[ V - \frac{g}{k} \ln(g + kV) \right]$$

$$H = \frac{1}{k} \left[ V - \frac{g}{k} \ln(g + kV) + \frac{g}{k} \ln g \right]$$

$$= \frac{1}{k} \left[ V - \frac{g}{k} \ln \left( \frac{g + kV}{g} \right) \right]$$

$$= \frac{V}{k} - \frac{g}{k^2} \ln \left( 1 + \frac{kV}{g} \right)$$

# QUESTION 6 (CONT)

c) i) roots are

$$\begin{aligned} z_1 &= 1 \\ z_2 &= \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7} & (= w) \\ z_3 &= \cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} & (= w^2) \\ z_4 &= \cos \frac{6\pi}{7} + i \sin \frac{6\pi}{7} & (= w^3) \\ z_5 &= \cos \frac{8\pi}{7} + i \sin \frac{8\pi}{7} & (= w^4) \\ z_6 &= \cos \frac{10\pi}{7} + i \sin \frac{10\pi}{7} & (= w^5) \\ z_7 &= \cos \frac{12\pi}{7} + i \sin \frac{12\pi}{7} & (= w^6) \end{aligned}$$

(or equivalent expressions)

ii) roots can be expressed as  $1, w, w^2, w^3, w^4, w^5, w^6$

$\therefore$  sum of roots  $(-\frac{b}{a})$  is

$$1 + w + w^2 + w^3 + w^4 + w^5 + w^6 = 0$$

iii) roots  $w + w^6, w^2 + w^5, w^3 + w^4$

Sum of roots 1 at a time:  $w + w^6 + w^2 + w^5 + w^3 + w^4 = -1$

$$\begin{aligned} \text{Sum of roots 2 at a time: } & (w + w^6)(w^2 + w^5) + (w + w^6)(w^3 + w^4) + \\ & (w^2 + w^5)(w^3 + w^4) \\ &= w^3 + w^4 + w^8 + w^9 + w^4 + w^3 + w^9 + w^{10} + w^5 + w^6 + w^8 + w^9 \\ &= 2(w + w^2 + w^3 + w^4 + w^5 + w^6) \\ &= -2 \end{aligned}$$

note  $w^8 = w^1$   
 $w^9 = w^2$   
etc.

Sum of roots 3 at a time:

$$\begin{aligned} & (w + w^6)(w^2 + w^5)(w^3 + w^4) \\ &= (w + w^6)(w^5 + w^6 + w^8 + w^9) \\ &= w^6 + w^7 + w^9 + w^{10} + w^{11} + w^{12} + w^{14} + w^{15} \\ &= w^6 + 1 + w^2 + w^3 + w^4 + w^5 + 1 + w \\ &= 2 + -1 \\ &= 1 \end{aligned}$$

$\therefore$  required polynomial

$$\begin{aligned} x^3 - (-1)x^2 + (-2)x - (1) &= 0 \\ x^3 + x^2 - 2x - 1 &= 0 \end{aligned}$$

QUESTION 7

a) i)  $\angle PQA = \theta$  (angle between tangent and chord equals angle in the alternate segment)

ii)  $\angle OBA = \theta$  (angles opposite equal sides of triangle are equal)  
( $OA = OB$  radii of circle)

$\therefore O, B, Q, A$  concyclic as  $OA$  subtends equal angles at  $B$  and  $Q$ .

iii)  $\angle BQO = \angle BAO = \theta$  (chord  $OB$  subtends equal angles at  $Q$  and  $A$ )  
( $O, B, Q, A$  concyclic)

$$\therefore \angle BQO = \angle PQA = \theta$$

$\therefore OQ$  bisects  $\angle BQA$ .

iv)  $\angle BRA = \angle OAB = \theta$  (angle between tangent and chord equals angle in alternate segment)

$$\therefore \angle BRA = \angle BQO = \theta$$

$\therefore OQ \parallel AR$  (corresponding angles equal)

b) i) each rectangle has width 1 unit and heights  $\ln 2, \ln 3, \ln 4, \dots, \ln k$  and there are  $(k-1)$  rectangles.

$$\begin{aligned} \therefore \text{sum of areas} &= 1 \times \ln 2 + 1 \times \ln 3 + 1 \times \ln 4 + \dots + 1 \times \ln k \\ &= \ln 2 + \ln 3 + \ln 4 + \dots + \ln k \end{aligned}$$

$$\begin{array}{lll} \text{ii)} & \int_2^{k+1} \ln(x-1) dx & u = \ln(x-1) \quad u' = \frac{1}{x-1} \\ & \text{using parts} & v = x \quad v' = 1 \end{array}$$

$$= x \ln(x-1) \Big|_2^{k+1} - \int_2^{k+1} \frac{x}{x-1} dx$$

$$= (k+1) \ln k - 2 \ln 1 - \left[ \int_2^{k+1} 1 + \frac{1}{x-1} dx \right]$$

$$\begin{aligned}
&= (k+1) \ln k - \left[ x + \ln(x-1) \right]_2^{k+1} \\
&= (k+1) \ln k - [(k+1) + \ln k] - (2 + \ln 1) \\
&= (k+1) \ln k - k - 1 - \ln k + 2 \\
&= k \ln k + \ln k - k - \ln k + 1 \\
&= k \ln k - k + 1
\end{aligned}$$

iii) from diagram we can see

$$\int_2^{k+1} \ln(x-1) dx < S < \int_2^{k+1} \ln x dx$$

$$\begin{aligned}
\text{now } \int_2^{k+1} \ln x dx &= x \ln x \Big|_2^{k+1} - \int_2^{k+1} dx \\
&= [(k+1) \ln(k+1) - 2 \ln 2] - [x]_2^{k+1} \\
&= (k+1) \ln(k+1) - \ln 4 - k + 1
\end{aligned}$$

$$\begin{aligned}
\text{and } S &= \ln 2 + \ln 3 + \ln 4 + \dots + \ln k \\
&= \ln k!
\end{aligned}$$

$$\begin{aligned}
\therefore k \ln k - k + 1 &< \ln k! < (k+1) \ln(k+1) - \ln 4 - k + 1 \\
k \ln k &< \ln k! + k - 1 < (k+1) \ln(k+1) - \ln 4
\end{aligned}$$

$$\ln k^k < \ln k! + k - 1 < \ln \left( \frac{k+1}{4} \right)^{k+1}$$

$$k^k < e^{\ln k! + k - 1} < \left( \frac{k+1}{4} \right)^{k+1}$$

$$k^k < k! e^{k-1} < \frac{1}{4} (k+1)^{k+1}$$

# QUESTION 8

a)  $\sin x = \cos 5x$

$$\cos\left(\frac{\pi}{2} - x\right) = \cos 5x$$

$$5x = 2n\pi \pm \left(\frac{\pi}{2} - x\right)$$

$$\therefore 5x = 2n\pi + \left(\frac{\pi}{2} - x\right)$$

$$5x = 2n\pi - \left(\frac{\pi}{2} - x\right)$$

$$x = \frac{1}{6} \left(2n\pi + \frac{\pi}{2}\right)$$

$$x = \frac{1}{4} \left(2n\pi - \frac{\pi}{2}\right)$$

when  $n=0$

$$x = \frac{\pi}{12}$$

$n=1$

$$x = \frac{3\pi}{8}$$

$n=1$

$$x = \frac{5\pi}{12}$$

$n=2$

$$x = \frac{7\pi}{8}$$

$n=2$

$$x = \frac{3\pi}{4}$$

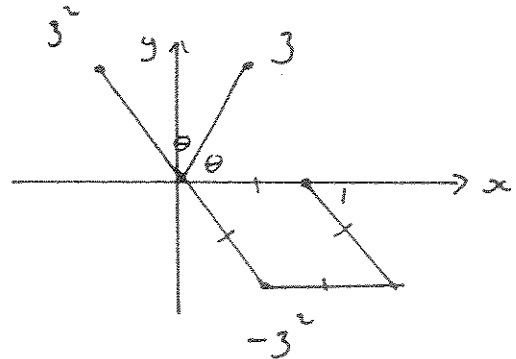
$$\therefore \text{Solutions } x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{3\pi}{8}, \frac{7\pi}{8}$$

b)  $\text{Arg}\left(\frac{2}{1-z^2}\right)$

$$= \text{Arg } 2 - \text{Arg}(1-z^2)$$

$$= 0 - \frac{1}{2}(\pi - 2\theta)$$

$$= \frac{\pi}{2} - \theta$$



c) i)  $-1 \leq f(x) \leq 1$

ii) a)  $x_1 = g(r)$

$$x_1 = \frac{e^{2r} - 1}{e^{2r} + 1}$$

$$x_1 e^{2r} + x_1 = e^{2r} - 1$$

$$e^{2r} = \frac{1 + x_1}{1 - x_1}$$

$$r = \frac{1}{2} \ln\left(\frac{1 + x_1}{1 - x_1}\right)$$

( $\beta$ )  $1 + x_1$  and  $1 - x_1$  are both positive as  $-1 < x_1 < 1$

$$\therefore \frac{1 + x_1}{1 - x_1} \text{ is positive}$$
$$\therefore \ln \left( \frac{1+x_1}{1-x_1} \right) \text{ exists}$$

$\therefore r$  exists

$$\begin{aligned}
 (8) \quad \frac{2g(x)}{1+(g(x))^2} &= \frac{2(e^{2x}-1)}{e^{2x}+1} \div \left(1 + \left(\frac{e^{2x}-1}{e^{2x}+1}\right)^2\right) \\
 &= \frac{2(e^{2x}-1)}{e^{2x}+1} \times \frac{(e^{2x}+1)^2}{(e^{2x}+1)^2 + (e^{2x}-1)^2} \\
 &= \frac{2(e^{2x}-1)(e^{2x}+1)}{2(e^{4x}+1)} \\
 &= \frac{e^{4x}-1}{e^{4x}+1} \\
 &= g(2x)
 \end{aligned}$$

(8) from part (b) the result is true for  $n=1$   
 assume true for  $n=k$  i.e.  $x_k = 9(2^{k-1} + 1)$   
 test for  $n=k+1$

$$\begin{aligned} x_{k+1} &= f(x_k) \\ &= \frac{2x_k}{1+(x_k)^2} \\ &= \frac{2g(2^{k-1}.r)}{1+(g(2^{k-1}.r))^2} \\ &= g(2.2^{k-1}.r) \\ &= g(2^k.r) \end{aligned}$$

which is the required result

which is the required result